6.5 The Indefinite Integral; Method of Substitution

Name _____ Period ____ Date _____

AP Calculus 6.5 · Student sheet

I can…

- find an indefinite integral using substitution.
- change the limits of integration when I substitute in a definite integral.
- use even and odd symmetry to evaluate integrals.

Vocabulary · fill in the missing word

Indefinite integral — $\int f(x)\,dx = F(x) + C$, all the _____ of f

Substitution — let $u = g(x)$, so $du = g'(x)\,dx$, and rewrite the _____ in u

Even function — $f(-x) =$ _____; then $\int$ from $-a$ to $a = 2\int$ from 0 to a

Odd function — $f(-x) =$ _____; then $\int$ from $-a$ to $a = 0$

Part 1 · Undo the Chain Rule

Differentiate, then reverse.

F(x)	F′(x)	So $\int F'(x)\,dx =$
$(x^2 + 1)^4$		
$\sin(3x)$		
$e^{(x^2)}$		

Part 2 · Notes and examples (fill in each note, then work the example)

Substitution: $\int f(g(x))g'(x)\,dx = \int f(u)$ _____, where $u = g(x)$.

Example 1

Find $\int 2x(x^2 + 1)^4\,dx$.

Adjust a constant: if $du = k\,dx$, then $dx = du/k$, so divide by the _____ k.

Example 2

Find $\int \cos(3x)\,dx$.

Example 3

Find $\int xe^{x^2}\,dx$.

Definite integrals: change the limits of integration to _____-values.

Example 4

Evaluate $\int_0^1 x(x^2 + 1)^3\,dx$.

Symmetry: for an odd function, $\int$ from $-a$ to a of $f(x)\,dx$ = _____.

Example 5

Evaluate $\int_{-2}^{2} x^3\,dx$ and $\int_{-2}^{2} x^2\,dx$.

Part 3 · Partner practice

Partner 1

Find $\int e^{4x}\,dx$.

Partner 2

(AP® Practice) $\int \frac{x}{1 + x^2}\,dx$ =

(A) $(\frac{1}{2})\ln(1 + x^2) + C$ (B) $\tan^{-1} x + C$

(C) $\ln(1 + x^2) + C$ (D) $2\ln(1 + x^2) + C$